

Lecture 1: Foundations of Behavioural Economics

Limits of Rationality — Why the Standard Model Breaks Down

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ECON 230 · Introduction to Behavioural Economics · Winter 2027

The Allais Paradox — Your Turn

Problem A — Choose ONE:

- Option 1: \$1,000,000 **for certain**
- Option 2: 89% chance of \$1M, 10% chance of \$5M, 1% chance of \$0

Problem B — Choose ONE:

- Option 1: 11% chance of \$1M, 89% chance of \$0
- Option 2: 10% chance of \$5M, 90% chance of \$0

Write your choices down. We will return to this in 30 minutes.

Foundations

1. Limits of Rationality ← Today
2. Heuristics and Biases
3. Prospect Theory

Risk & Time

4. Time Inconsistency & Present Bias
5. Savings Behaviour

Social & Strategic

6. Social Preferences
7. Behavioural Game Theory

Markets & Policy

8. Behavioural Finance
9. Nudges & Choice Architecture
10. Applications & Policy Design

What Does Economics Assume?

The Rational Agent

Standard economics assumes agents are: **fully rational, perfectly informed, and purely self-interested.**

- **Complete preferences:** You can rank *every* possible outcome. No indecision.
- **Transitivity:** If you prefer A to B and B to C, you prefer A to C. Always.
- **Full optimisation:** Given all information, you compute the globally optimal action — at zero cognitive cost.
- **Bayesian updating:** When new information arrives, you update your beliefs exactly as Bayes' rule dictates.
- **Time consistency:** Your preferences do not change as options approach in time.

Why assume this?

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Expected Utility Theory

Von Neumann & Morgenstern (1944)

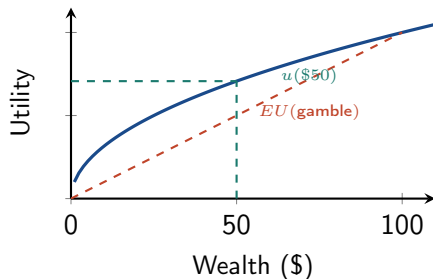
Under uncertainty, a rational agent maximises:

$$EU = \sum_i p_i \cdot u(x_i)$$

where p_i is the probability of outcome x_i and $u(\cdot)$ is a utility function.

- **Example:** Gamble A = 50% chance of \$100, 50% chance of \$0.
- $EU(A) = 0.5 \cdot u(\$100) + 0.5 \cdot u(\$0)$
- A concave u captures risk aversion: prefer \$50 for sure over the gamble.
- Rests on 4 axioms: completeness, transitivity, continuity, independence.
- Violate any axiom $\Rightarrow$ EU collapses.

Concave Utility: Risk Aversion



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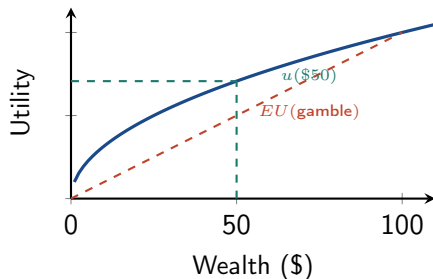
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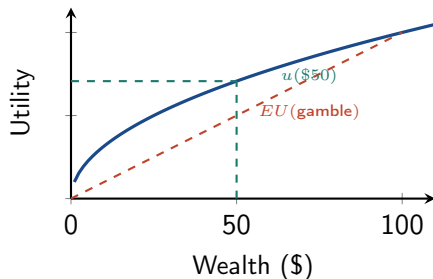
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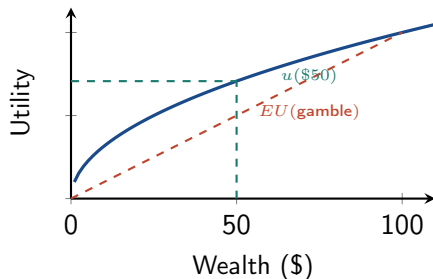
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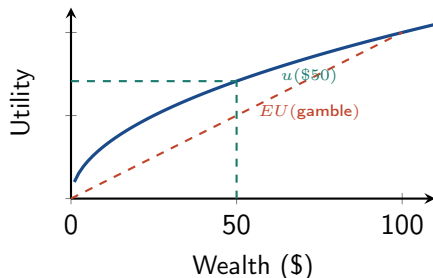
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The Four Axioms in Plain English

Completeness

You can always compare two options. “I just can’t decide” is not allowed.

Transitivity

If $A \succ B$ and $B \succ C$, then $A \succ C$. No preference cycles.

Why it matters: Independence makes probabilities “multiply through” in EU — and it is precisely what Allais showed **most people violate systematically**.

Continuity

Small changes in probability create small changes in preferences. No “lexicographic” preferences.

Independence (The Critical Axiom)

If $A \succ B$, mixing with C preserves:
 $(pA + (1 - p)C) \succ (pB + (1 - p)C)$.

The Power of the Rational Model

- **Nuclear deterrence:** Schelling's (1960) rational-actor analysis of mutually assured destruction.
- **Portfolio theory:** Markowitz (1952) — diversification as rational risk management. *Nobel 1990.*
- **Auction design:** Milgrom & Wilson (2020) — FCC spectrum auctions earned billions for US taxpayers using rational bidding theory.
- **Insurance markets:** Arrow (1963) — why insurance exists and when it fails.
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The Point

Do NOT dismiss the rational model. It is enormously powerful. Our goal is to understand **where it breaks down** — and what replaces it there.

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The Allais Paradox — Back to Your Choices

What Most People Choose

- Problem A: **Option 1** (certain \$1M)
- Problem B: **Option 2** (10% chance of \$5M)

The Problem

Under EU, let $u(0) = 0$. Then:

$$\begin{aligned} A1 \succ A2 &\Rightarrow u(1M) > 0.89 u(1M) + 0.1 u(5M) \\ &\Rightarrow 0.11 u(1M) > 0.1 u(5M) \end{aligned}$$

But $B2 \succ B1$ implies the **opposite!**

Contradiction.

Allais (1953)

Maurice Allais presented this to leading economists at a Paris conference. **Paul Samuelson** (Nobel 1970) initially violated it himself before correcting his answer. “The Allais Paradox” has been replicated worldwide across cultures, stake sizes, and expert audiences.

The violation of the Independence Axiom is not a mistake — it is a systematic preference.

The Ellsberg Paradox (1961)

The Setup

An urn contains 90 balls: 30 are red; the remaining 60 are either black or yellow in unknown proportions.

Problem 1: Bet on...

- A: Win \$100 if red drawn
- B: Win \$100 if black drawn

Most prefer A.

Problem 2: Bet on...

- C: Win \$100 if red or yellow
- D: Win \$100 if black or yellow

Most prefer D.

The Contradiction

$A \succ B$ implies $P(\text{red}) > P(\text{black})$. But
 $D \succ C$ implies $P(\text{black}) > P(\text{red})$.

Violates Savage's Sure-Thing Principle.

Key Insight

People dislike **ambiguity** (unknown probabilities) more than **risk** (known probabilities). This is **ambiguity aversion** — distinct from risk aversion.

Gain Frame

A disease will kill 600 people. Choose:

- A: 200 people will be **saved**
- B: 1/3 prob all 600 saved; 2/3 prob nobody saved

72% choose A.

Loss Frame

Same disease. Choose:

- C: 400 people will **die**
- D: 1/3 prob nobody dies; 2/3 prob 600 die

78% choose D.

A = C and B = D — Mathematically Identical

Under EU, the same person should choose either $\{A, C\}$ or $\{B, D\}$. Yet most choose $\{A, D\}$.

Kahneman & Tversky, 1981, Science. Replicated in 50+ countries.

Loss Aversion — The Asymmetry of Pain

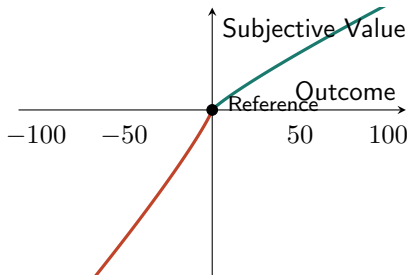
The Finding

Losses loom approximately 2–2.5 times larger than equivalent gains.

$\lambda \approx 2.25$ (Tversky & Kahneman, 1992)

- Ask: “Would you accept a 50/50 bet to win \$150 or lose \$100?” Most say **no**.
- EU with a concave function cannot explain this for small stakes.
- Consumer panel study (butter): $\lambda \approx 2.4$ (Hardie, Johnson & Fader, 1993).
- Labour market: workers resist nominal wage cuts even when real wages are falling (Akerlof et al., 1996).

S-Shaped Value Function (Preview)



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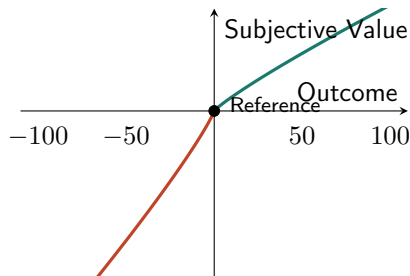
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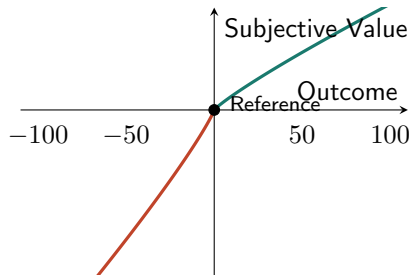
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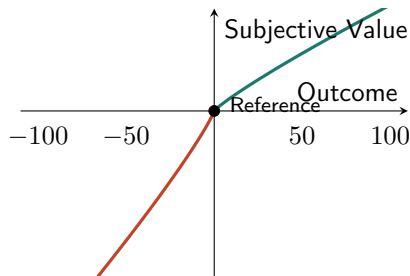
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Thaler (1981) — The Declining Discount Rate

Participants asked: “What amount would make you indifferent between receiving it now vs. later?”

	Delay: 1 month	Delay: 1 year	Delay: 10 years
\$15 now → \$X later	\$20	\$50	\$100
<i>Implied annual rate</i>	<i>345%</i>	<i>120%</i>	<i>19%</i>

The Problem

Standard exponential discounting predicts a **constant** annual discount rate. But observed rates **decline sharply** with the delay horizon. This generates **time inconsistency**: preferences reverse as the future becomes the present.

Consequence: “I’ll start saving next month” — said every month for 20 years.

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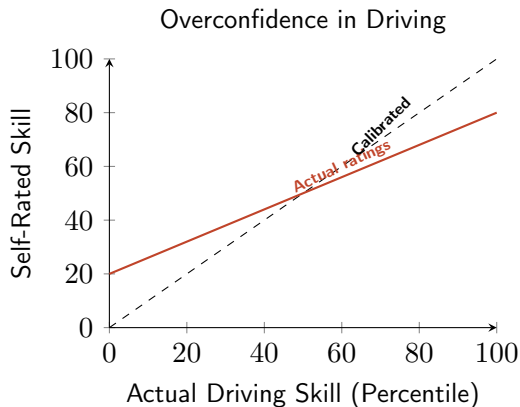
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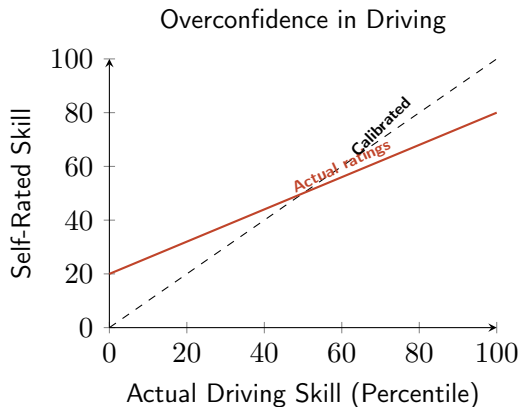
Overconfidence — The Evidence

- **Driving:** 93% of US drivers rate themselves “above average” (Svenson, 1981). *Impossible by definition.*
- **Teaching:** 94% of professors at one US university rated themselves above-average teachers (Cross, 1977).
- **Exams:** Students predict 72% average; actual = 67% (Dunning et al., 2003).
- **Investments:** Men trade 45% more than women; earn 1.4% less/year (Barber & Odean, 2001).
- **Projects:** Planning fallacy — Sydney Opera House budgeted \$7M, cost \$102M.



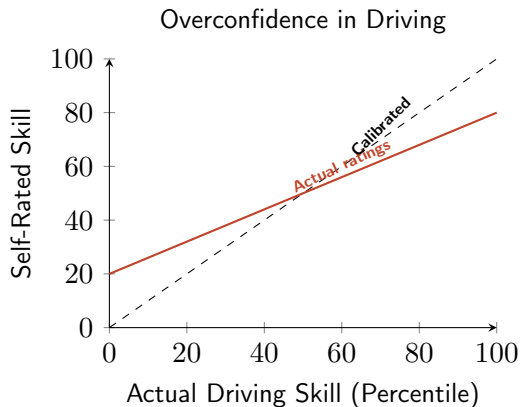
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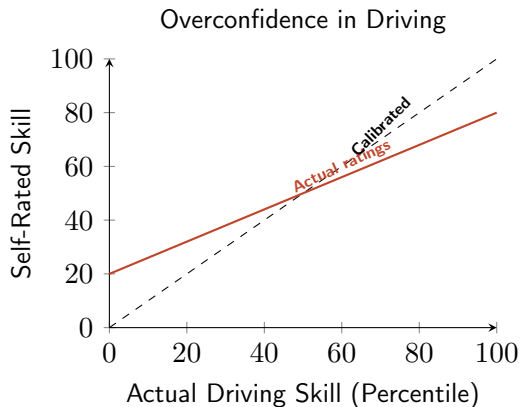
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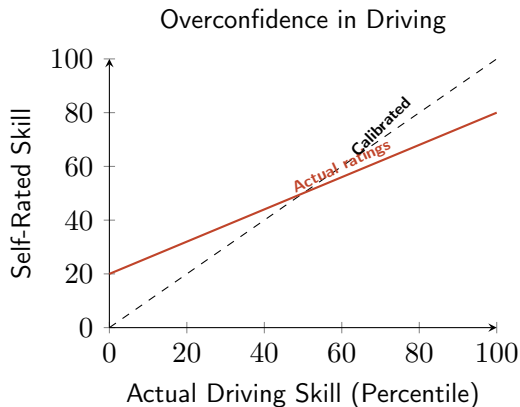
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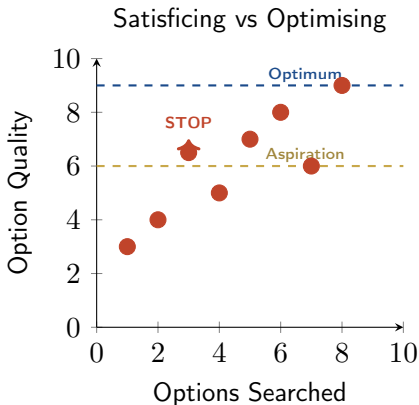
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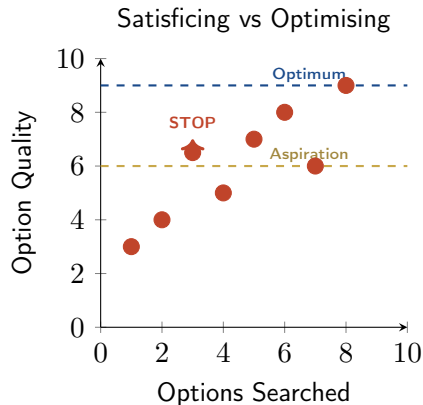
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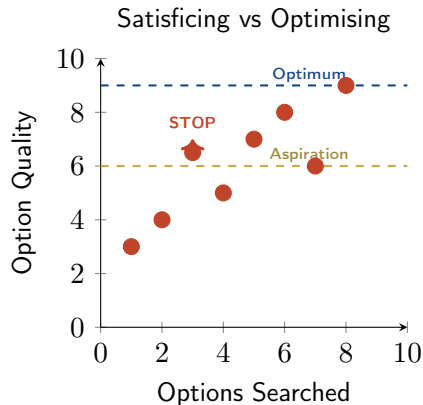
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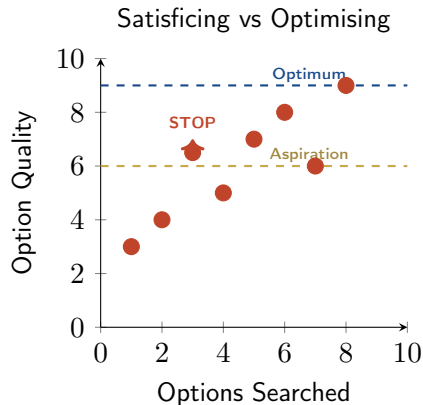
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Simon (1955)

“A wealth of information creates a poverty of attention.”

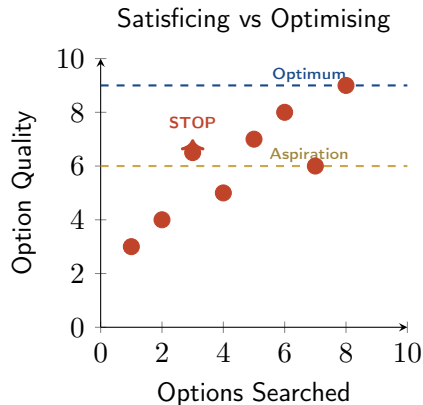
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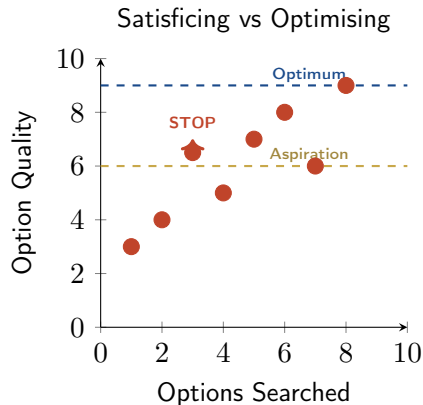
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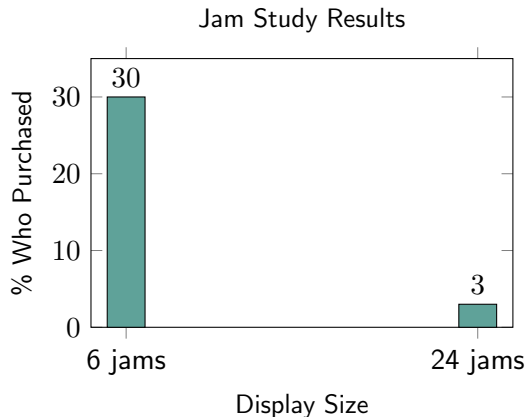
The Paradox of Choice

The Jam Study (Iyengar & Lepper, 2000)

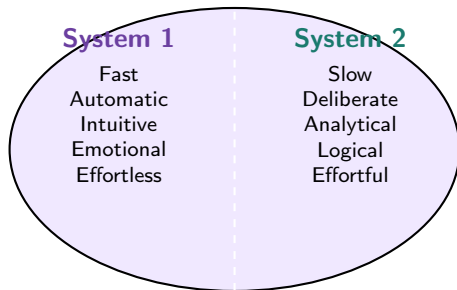
- **24-jam display:** 60% of shoppers stopped; **3%** bought.
- **6-jam display:** 40% stopped; **30%** bought.
- *More choice $\Rightarrow$ less action and less satisfaction.*

Canadian Context

Choi et al. (2009): Canadians offered **more mutual fund options** in their RRSP contribute **less** and choose **worse** options. Choice overload is real in Canadian retirement savings.



System 1 vs System 2 (Kahneman, 2011)



The Bat-and-Ball Problem

A bat and ball cost \$1.10 together. The bat costs \$1.00 more than the ball. How much does the ball cost?

System 1 says: \$0.10. **Wrong.**

Correct answer: \$0.05.

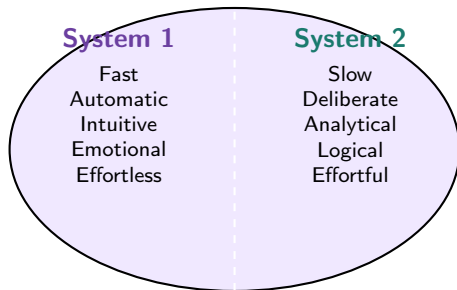
If ball = \$0.10, bat = \$1.10, total = \$1.20.

Contradiction.

90% of people give the wrong answer.

Even Harvard, MIT and Princeton students:
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[GAME] Cognitive Reflection Test

Rules: 2 minutes, write your answers now, no calculators

Q1. A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost?

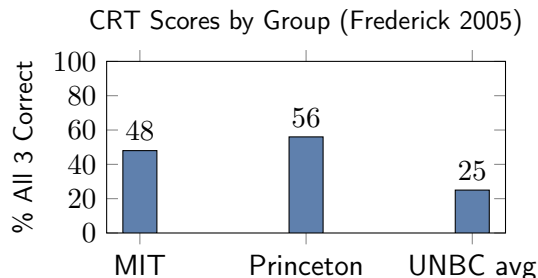
Q2. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?

Q3. In a lake, a patch of lily pads doubles in size every day. After 48 days the lake is half full. After how many days is the lake completely full?

Answers revealed on next slide...

Answers

- Q1: \$0.05 (*not \$0.10*)
- Q2: 5 minutes (*not 100*)
- Q3: 49 days (*not 24*)

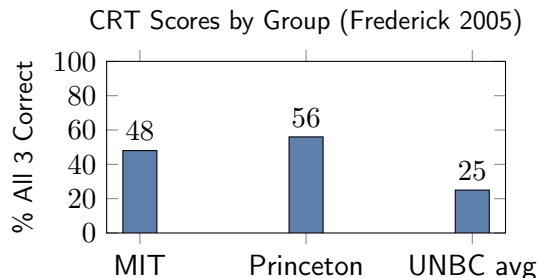


- Only **17%** of US college students get all 3 correct (Frederick, 2005).
- MIT: 48%. Princeton: 56%.
- **System 1 hijacks System 2** even for intelligent people under time pressure.
- Low CRT scores predict: higher present bias, more overconfidence, lower savings rates, poorer financial literacy.
- *This is not about intelligence. It's about cognitive engagement.*

CRT Results and Discussion

Answers

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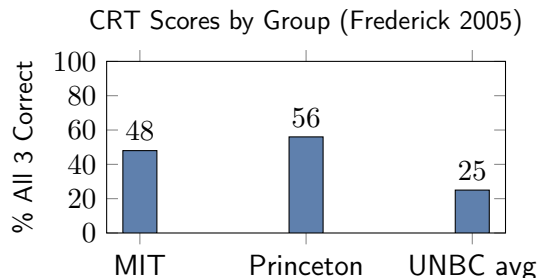


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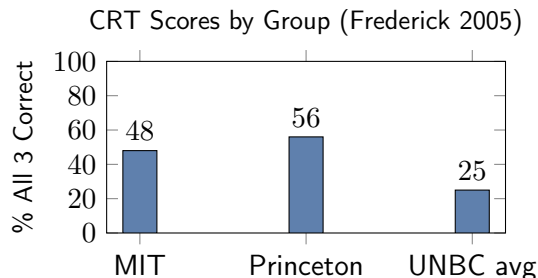
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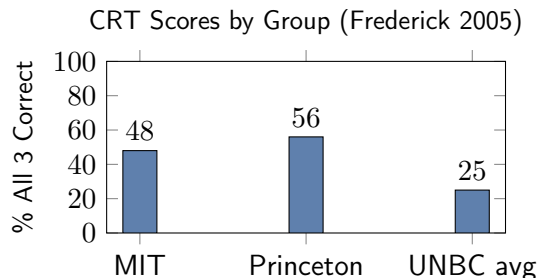
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Group A (Left Side)

Is the population of Canada **more or less than 10 million**? What is your best estimate of Canada's population?

Group B (Right Side)

Is the population of Canada **more or less than 100 million**? What is your best estimate of Canada's population?

Expected Results (Tversky & Kahneman 1974 design)

- Group A mean estimate: $\approx$ 22–28 million
- Group B mean estimate: $\approx$ 40–55 million
- **Correct answer: approximately 40.5 million (Statistics Canada, 2024)**
- The **irrelevant** anchor number shifts estimates by 30–50%.

Markets Do Not Fix Irrational Behaviour

- **Traditional argument:** Markets select against irrational agents — they lose money and exit.
- **Counter-argument 1 (DeLong et al., 1990):** Limits of arbitrage. Rational traders cannot always profit from others' mistakes — mispricing can persist or worsen before correcting.
- **Counter-argument 2:** When errors are **correlated** (everyone makes the same mistake), they don't cancel — they amplify. Example: 2008 financial crisis — sophisticated institutions all made the same overconfidence mistake simultaneously.
- **Counter-argument 3:** Noise traders can *earn higher expected returns* than rational arbitrageurs if their sentiment risk is priced (DeLong et al., 1990).

Bottom Line

Markets reduce but do not eliminate the consequences of systematic cognitive errors.

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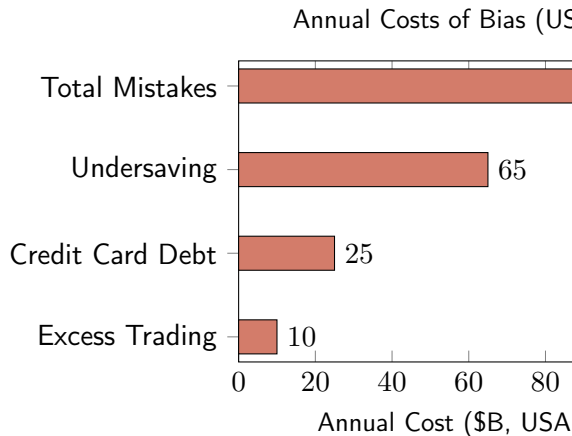
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The Cost of Cognitive Bias — Real Data

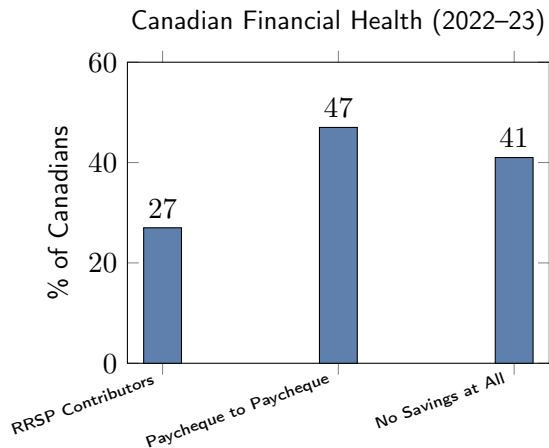
- **\$80–100 billion:** Annual cost to US households from investment mistakes (Choi et al., 2010).
- **75%:** Share of credit card debt explained by present bias (Laibson et al., 2001).
- **1.4%/year:** Lower net returns from male overconfidence in trading (Barber & Odean, 2001).
- **27%:** Canadians who contributed to RRSPs in 2022 — despite average unused room of \$25,000+ (Statistics Canada).



Canadian Savings Data (Statistics Canada, 2022–2024)

- Only **27%** of eligible Canadians contributed to RRSPs in 2022.
- Median unused RRSP room: **\$25,000+** per filer.
- **47%** of Canadians live paycheque-to-paycheque (FP Canada, 2023).
- Average credit card debt: **\$4,200** at 19.99% APR (Equifax, 2023).

93% of Canadians *know* RRSPs exist. The problem is not information — it is **present bias**.



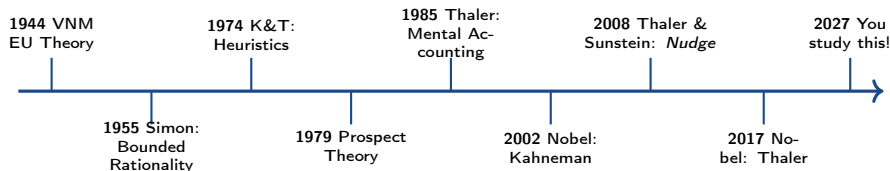
Three Approaches to Economic Policy

Approach	Assumption	Policy Tool	Example
Standard Econ	Full rationality	Prices, information	Carbon tax, calorie labels
Behavioural	Bounded rationality	Nudges, defaults, commitment	Auto-enrolment, SMarT
Hard Paternalism	Irrationality	Mandates, bans	Helmet laws, sugar bans

The Behavioural Insight

Behavioural policy does not replace standard policy — it **complements** it by working with cognitive architecture, not against it.

A Brief History of Behavioural Economics



Key Figures

Herbert Simon (Nobel 1978) · **Daniel Kahneman** (Nobel 2002) · **Vernon Smith** (Nobel 2002) · **Richard Thaler** (Nobel 2017)

- **Laboratory experiments:** Controlled environment, randomisation, replication. Trade-off: external validity.
- **Field experiments:** Real-world randomised controlled trials (RCTs). “Nudge units” run these in governments.
- **Natural experiments:** Exploit exogenous variation. Organ donation law changes, pension regulation changes.
- **Surveys:** Elicit stated preferences, beliefs, and time preferences. Canadian Survey of Financial Security (Statistics Canada).
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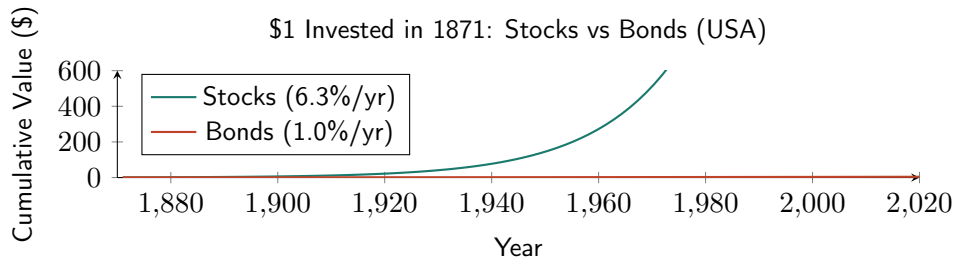
The Equity Premium Puzzle — Preview

Mehra & Prescott (1985)

US stocks have returned $\approx 6\%$ more per year than bonds since 1871. To justify this within the EU framework, the coefficient of relative risk aversion would need to be $\gamma > 30$ — *implausibly high*.

Behavioural Explanation (Benartzi & Thaler, 1995)

If investors are loss averse ($\lambda = 2.25$) and evaluate their portfolio **annually**, the required equity premium is exactly $\approx 6\%$. The puzzle dissolves with myopic loss aversion.



Discussion Questions

- Q1. “Is it fair to call someone *irrational* when they make a systematic mistake? Who decides what the *right* answer is?”
- Q2. “If people aren’t fully rational, does this mean governments *should* decide for them? Or does government face the same cognitive limits?”
- Q3. “Can you think of a decision you made recently that might have violated transitivity, independence, or consistency? Did you feel irrational at the time?”

Think–Pair–Share (3 minutes)

Discuss Q1 with a neighbour. One person will be asked to share.

Key Takeaways — Lecture 1

- ✓ **Standard model:** rational agents maximise expected utility; four axioms underpin it.
- ✓ **Evidence against:** Allais paradox (independence), Ellsberg (ambiguity), framing effects, loss aversion, overconfidence.
- ✓ **Bounded rationality (Simon):** people intend to be rational but are cognitively, informationally, and temporally constrained.
- ✓ **Satisficing:** search until “good enough”; heuristics substitute for full optimisation.
- ✓ **Two systems:** fast, intuitive System 1 often overrides deliberate System 2.
- ✓ **Errors are systematic and predictable** $\Rightarrow$ economically important $\Rightarrow$ addressable by policy.

The Central Insight of This Course

People are not irrational. They are *intentionally rational* — but cognitively constrained. Understanding those constraints is where behavioural economics begins.

Further Reading

- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus & Giroux. **Essential.**
- Thaler, R. & Sunstein, C. (2008). *Nudge*. Yale University Press.
- Ariely, D. (2008). *Predictably Irrational*. HarperCollins.
- Camerer, C., Loewenstein, G. & Rabin, M. (2004). *Advances in Behavioral Economics*. Princeton University Press. **Academic reference.**
- Tversky, A. & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185, 1124–1131.
- Von Neumann, J. & Morgenstern, O. (1944). *Theory of Games and Economic Behavior*. Princeton.

Next: Lecture 2: Heuristics and Biases — We drill deeper into the specific mental shortcuts that cause systematic errors: representativeness, availability, anchoring, confirmation bias, and overconfidence.